Fiscal Year:	FY 2005	**Task Last Updated:**	FY 04/03/2006
PI Name:	Rabin, Bernard M. Ph.D.		
Project Title:	Behavioral And Neurochemical Effects Of Exposure To Heavy Particles		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rabin@umbc.edu	**Fax:**	FY (410) 455-1055
PI Organization Type:	UNIVERSITY	**Phone:**	(410) 952-1761
Organization Name:	University of Maryland, Baltimore County		
PI Address 1:	Department of Psychology		
PI Address 2:	1000 Hilltop Cir		
PI Web Page:			
City:	Baltimore	**State:**	MD
Zip Code:	21250-0001	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	02-OBPR-02
Start Date:	06/15/2003	**End Date:**	06/14/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Shukitt-Hale, Barbara (Tufts University) Joseph, James A. (Tufts University)		
Grant/Contract No.:	NAG9-1529		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Exposing rats to non-lethal doses of 56Fe particles produces changes in the central nervous system, particularly in dopaminergic function. Dopamine is a neurotransmitter that mediates a wide range of behaviors, including motor behavior, some types of learning, reinforcement, sensory information processing, food intake and the production of abnormal behavior patterns. Because changes in each of these behaviors can affect the performance of astronauts, the knowledge of the behaviors affected by exposure and an understanding of the neural mechanisms mediating the behaviors is essential. The proposed research program is designed to gain additional information on alterations in brain neurochemistry and behavior following exposure to particles of high energy and charge, such as those that might be encountered during space travel outside the magnetic field of the earth. The research is designed to further describe and characterize behavioral and neurochemical deficits produced by exposure to heavy particles, determine their underlying mechanisms, and develop approaches to minimize such deficits. In addition, the research program proposes to examine		

	factors such as age and diet that may interact with the effects of exposure to heavy particles, and which may, in turn, affect the health and performance of astronauts during exploratory missions to other planets.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research which we have conducted previously has shown that exposing young rats (less than 3 mo. old) to low doses of heavy particles (56Fe, 600 MeV/n) has shown that the neurochemical deficits produced by this exposure are similar to those that are observed in aged rats (24 mo. old). The research which we have completed at BNL indicates that similar deficits are observed following exposure to one GeV/n 56Fe particles as well. Thus, exposing rats to low doses of 56Fe particles provides a way to accelerate aging in experimental animals with respect to certain brain and behavioral parameters so that experimenters do no have to wait for 24 months to obtain old animals. The research program is designed to understand the mechanisms by which exposure to heavy particles (primarily 56Fe) produce their effects on brain and behavior. Although the impetus for the research is to understand and minimize the effects of exposure to heavy particles on astronauts, the research program necessarily has implications for the understanding of the natural aging process. Because exposure to 56Fe particles may produce accelerated aging, this research is also indirectly concerned with the basic processes underlying the biology of aging. Similarly, because the ultimate goal of the research program is to develop interventions to minimize the effects of exposure to heavy particles on astronauts, the interventions may also prove useful with the natural aging process.
Task Progress:	Results indicate that there is a significant risk that exposure to low doses of HZE (high-energy radiation) particles could produce alterations in central nervous system (CNS) function and in the behaviors that are dependent upon the integrity of specific CNS systems. Effects of Diet on the Radiation-Induced Disruption of Behavioral and Neural Functioning: These results indicate that diets containing antioxidant phytochemicals can protect the organism against the deleterious effects of exposure to HZE radiation on neurochemical and behavioral endpoints. These results are consistent with the observation that exposure to ionizing radiation causes oxidative stress and the production of reactive oxygen species. However, the fact that the effectiveness of specific diets can vary as a function of the specific endpoint suggests that oxidative stress may be only one of the mechanisms underlying the HZE-particle induced disruption of neurochemical and behavioral function. Additional research with a variety of diets and a variety of endpoints will be needed to further delineate the underlying mechanisms. Effects of Age and Radiation on Behavior: In experiments completed last year, the results confirmed that the responsiveness of the organism to exposure to 56Fe particles does vary as a function of age. However, the relationship is complex, varying both as a function of the age of the organism at the time of irradiation and as a function of the specific behavioral endpoint. Results suggest that the relative effectiveness of exposure to different particles may vary as a function of both the particular particle and the specific endpoint under consideration. The data obtained thus far would suggest that it may not be possible to determine the acute effects of exposure to the range of HZE particles to which astronauts may be exposed on exploratory class missions by simply studying the effects of exposure to a single particle. Additional research will be needed to determine the nature of the relationship between exposure to different heavy particles and exposure-induced changes in neurobehavioral endpoints.
Bibliography Type:	Description: (Last Updated: 05/08/2025)
Abstracts for Journals and Proceedings	Rabin BM, Shukitt-Hale B, Joseph J, Todd P. "Diet as a factor in behavioral radiation protection following exposure to heavy particles." 20th Annual Meeting of the American Society for Gravitational and Space Biology, New York, NY, 9-13 Nov 2004 Gravit Space Biol Bull. 2004 Nov;18(1):26. , Nov-2004
Abstracts for Journals and Proceedings	Rabin, B. M., Shukitt-Hale, B., Joseph, J. A., Carey, A., Carrihill, K. L. "The effects of exposure to heavy particles on behavior of rats irradiated at different ages." 34th Meeting of the Society for Neuroscience, San Diego, CA, 23-27 Oct 2004. Program No, 896.9. Abstract Viewer/Itinerary Planner. Washington, DC: Society for Neuroscience, 2004. , Oct-2004
Articles in Peer-reviewed Journals	Rabin BM, Joseph JA, Shukitt-Hale B. "Effects of age and diet on the heavy particle-induced disruption of operant responding produced by a ground-based model for exposure to cosmic rays." Brain Res. 2005 Mar 2;1036(1-2):122-9. PMID: 15725409 , Mar-2005
Articles in Peer-reviewed Journals	Rabin BM, Joseph JA, Shukitt-Hale B. "Heavy particle irradiation, neurochemistry and behavior: thresholds, dose-response curves and recovery of function." Adv Space Res. 2004;33(8):1330-3. Review. PMID: 15803623 , Jun-2004
Articles in Peer-reviewed Journals	Shukitt-Hale B, Szprengiel A, Pluhar J, Rabin BM, Joseph JA. "The effects of proton exposure on neurochemistry and behavior." Adv Space Res. 2004;33(8):1334-9. PMID: 15803624 , Jun-2004
Articles in Peer-reviewed Journals	Casadesus G, Shukitt-Hale B, Cantuti-Castelvetri I, Rabin BM, Joseph JA. "The effects of heavy particle irradiation on exploration and response to environmental change." Adv Space Res. 2004;33(8):1340-6. PMID: 15803625 , Jun-2004
Articles in Peer-reviewed Journals	Rabin BM, Carrihill-Knoll KL, Carey A, Shukitt-Hale B, Joseph JA. "Effect of diet on the disruption of operant responding at different ages following exposure to 56Fe particles." Age. 2005 Mar;27(1):69-73. , Mar-2005
Articles in Peer-reviewed Journals	Casadesus G, Shukitt-Hale B, Stellwagen HM, Smith MA, Rabin BM, Joseph JA. "Hippocampal neurogenesis and PSA-NCAM expression following exposure to 56Fe particles mimics that seen during aging in rats." Exp Gerontol. 2005 Mar;40(3):249-54. PMID: 15763403 , Mar-2005
Articles in Peer-reviewed Journals	Rabin BM, Joseph JA, Shukitt-Hale B. "A longitudinal study of operant responding in rats irradiated when 2 months old." Radiat Research, in press, 2005. , Apr-2005

Articles in Peer-reviewed Journals	Rabin BM, Shukitt-Hale B, Joseph J, Todd P. "Diet as a factor in behavioral radiation protection following exposure to heavy particles." Gravit Space Biol Bull., in press, 2005. , Apr-2005
Presentation	Rabin, B. M., Joseph, J. A., Shukitt-hale, B, Carrihill-Knoll, K. L., Carey, A., & Maloney, B. "Individual differences in the response to 56fe particles: effect of age." Presented at the Bioastronautics Workshop, Galveston, TX, 10-12 Jan 2005. Jan-2005
Presentation	Rabin, B. M., Shukitt-Hale, B., Joseph, J. A., Carey, A., & Carrihill, K. L. "The effects of exposure to heavy particles on behavior of rats irradiated at different ages." Presented at the 34th Meeting of the Society for Neuroscience, San Diego, CA, 23-27 Oct 2004. Oct-2004
Presentation	Rabin, B. M., Shukitt-Hale, B., Joseph, J. A. & Carey, A. "Comparison of the effects of exposure to different particles or energies on behavioral responding in rats." Presented at the 35th Scientific Assembly of the Committee on Space Research (COSPAR), Paris, France, 18-25, July, 2004. Jul-2004
Presentation	Shukitt-Hale, B., Casadesus, G., Carey, A., Rabin, B. M., & Joseph, J. A. "Exposure to radiation accelerates normal brain aging and produces deficits in spatial learning and memory." Presented at the 35th Scientific Assembly of the Committee on Space Research (COSPAR), Paris, France, 18-25 July, 2004. Jul-2004
Presentation	Joseph, J. A., Shukitt-Hale, B., Carey, A, & Rabin, B. M. "Strawberry and blueberry supplementation may protect against increased oxidative stress vulnerability from both irradiation and aging." Presented at the 35th Scientific Assembly of the Committee on Space Research (COSPAR), Paris, France, 18-25 July, 2004. Jul-2004
Presentation	Rabin, B. M., Shukitt-Hale, B., Joseph, J. A., & Todd, P. "Diet as a factor in behavioral radiation protection following exposure to heavy particles." Presented at the 20th Annual Meeting of the American Society for Gravitational and Space Biology, New York, NY, 9-13 Nov, 2004. Nov-2004